A Markov-Chain Multi-Market Trading Strategy: Methodology, Implementation, and Backtest Design

September 27, 2026

Abstract

This note documents a discrete-state Markov-chain trading strategy that is applied across five global equity index universes (S&P 500, Hang Seng, Shanghai Composite, FTSE 100, Nikkei 225) and a fixed global macro basket (crypto, commodities, and leveraged/sector ETFs). Daily returns are z-scored and mapped onto a five-state alphabet, a train-only transition matrix is estimated per ticker, and tickers are retained only if their one-step-ahead directional signal clears a minimum edge and hit-rate threshold on the training data. Retained tickers generate Kelly-capped, fixed-holding-period trade signals on held-out test data, which are then run through a daily event-driven backtest with transaction costs, intraday stop losses, and a volatility/trend-based index-rotation fallback for days without an active signal. We describe the full pipeline, the risk management rules, the performance attribution methodology (Sharpe ratio and CAPM alpha/beta against SPY), and discuss the backtest's principal limitations, most notably survivorship bias in the ticker universes.

Contents

1 Motivation **2**

2 State Space and Transition Matrix Estimation **2**
2.1 Return discretization . . . 2
2.2 Transition matrix estimation with Laplace smoothing . . . 2

3 Edge Filtering **3**

4 Signal Generation and Position Sizing **3**
4.1 Daily capital allocation . . . 3

5 Risk Management **4**
5.1 Intraday stop losses . . . 4
5.2 Transaction costs . . . 4
5.3 Quiet-day index rotation fallback . . . 4

6 Backtest Engine **4**

7 Performance Attribution **5**

8 Discussion and Limitations **5**

1 Motivation

Momentum and mean-reversion effects in daily equity returns are well documented empirically, but their sign and strength are regime- and asset-dependent. Rather than assume a single global rule (e.g. "buy after two down days"), this project estimates a *per-ticker*, data-driven transition structure: each ticker's own history is discretized into a small state alphabet describing the size of its daily move, and a transition matrix over that alphabet is fit on a training sample. If the fitted matrix implies a strong enough directional edge out-of-sample, the ticker is included in the tradable universe; if not, it is discarded. This turns "does momentum/mean-reversion exist here" into an explicit, per-ticker statistical test with a pre-registered threshold, rather than a strategy applied uniformly to every instrument.

2 State Space and Transition Matrix Estimation

2.1 Return discretization

For a given ticker, let r_t denote the daily percentage return computed from adjusted close prices. The full return history is split chronologically into a training segment (the first 70% of observations) and a held-out test segment (the remaining 30%):

$$r_1, \ldots, r_T \;=\; \underbrace{r_1, \ldots, r_{\lfloor 0.7T \rfloor}}_{\text{train}} \;\cup\; \underbrace{r_{\lfloor 0.7T \rfloor + 1}, \ldots, r_T}_{\text{test}}.$$

The training-set mean μ and (population) standard deviation σ are computed *only* from the training segment, and used to z-score *both* segments:

$$z_t = \frac{r_t - \mu}{\sigma}, \qquad t = 1, \ldots, T.$$

Using train-only statistics to standardize the test segment avoids look-ahead bias: nothing about the test-period distribution of returns influences the discretization boundaries.

Each z_t is then mapped to one of five states using two boundary parameters, $\delta = 0.55$ (the "moderate move" boundary) and $\varepsilon = 1.15$ (the "extreme move" boundary):

$$s_t = \begin{cases} \text{Very Down} & z_t \le -\varepsilon \\ \text{Down} & -\varepsilon < z_t \le -\delta \\ \text{Neutral} & -\delta < z_t < \delta \\ \text{Up} & \delta \le z_t < \varepsilon \\ \text{Very Up} & z_t \ge \varepsilon \end{cases}$$

2.2 Transition matrix estimation with Laplace smoothing

Let $\mathcal{S} = \{\text{Very Down}, \text{Down}, \text{Neutral}, \text{Up}, \text{Very Up}\}$. From the training-set state sequence $s_1, \ldots, s_n$, raw transition counts are accumulated:

$$N_{ij} = \sum_{t=1}^{n-1} \mathbf{1}\{s_t = i,\, s_{t+1} = j\}, \qquad i, j \in \mathcal{S}.$$

To avoid zero-probability transitions for state pairs that are rare or absent in the training window, Laplace (additive) smoothing with constant $\alpha = 0.7$ is applied before row-normalizing:

$$\hat{P}(j \mid i) = \frac{N_{ij} + \alpha}{\sum_{k \in \mathcal{S}} (N_{ik} + \alpha)}.$$

This yields a 5×5 row-stochastic transition matrix $\hat{P}$, estimated entirely from the training segment.

3 Edge Filtering

Not every ticker's transition matrix implies a usable trading edge. For each state i, define the *net directional probability*

$$\Delta(i) = \big[\hat{P}(\text{Up} \mid i) + \hat{P}(\text{Very Up} \mid i)\big] - \big[\hat{P}(\text{Down} \mid i) + \hat{P}(\text{Very Down} \mid i)\big].$$

A ticker is tested for edge by walking the training-set state sequence: at each step t where $|\Delta(s_t)| \geq \tau$ (edge threshold, $\tau = 0.2$), a directional prediction is made, $\text{sign}(\Delta(s_t)) \in \{+1, -1\}$, and scored against the *realized* next-day return r_{t+1} as a hit if $\text{sign}(\Delta(s_t)) \cdot r_{t+1} > 0$.

Let H be the number of such qualifying predictions and h the number of hits. The ticker clears the edge filter only if:

- $H \geq 50$ (minimum sample-size guard against spurious small-sample edges), and
- $h/H \geq 0.55$ (minimum hit rate).

Tickers that fail either condition are dropped from the tradable universe entirely — no signals are generated from them on the test set. This is an in-sample (training-set) filter; whether the edge *persists* out-of-sample is exactly what the test-set backtest evaluates.

4 Signal Generation and Position Sizing

For each ticker that clears the edge filter, its transition matrix $\hat{P}$ is applied to the **held-out test-set** state sequence to generate trade signals. At test-set time t with state s_t:

$$\Delta_{\text{long}} = \max(\Delta(s_t), 0), \qquad \Delta_{\text{short}} = \max(-\Delta(s_t), 0).$$

A signal fires only if $\max(\Delta_{\text{long}}, \Delta_{\text{short}}) \geq \tau$. The trade direction is Long if $\Delta_{\text{long}} \geq \Delta_{\text{short}}$, else Short. The position's raw Kelly-style weight is

$$\kappa = \min\big(\max(\Delta_{\text{long}}, \Delta_{\text{short}}),\ \kappa_{\max}\big), \qquad \kappa_{\max} = 0.20,$$

i.e. the magnitude of the directional edge itself is used as a (capped) sizing signal — larger edges get proportionally larger size, up to a 20% cap. Each signal has a fixed holding period of $H_d = 5$ trading days: the position is entered at the test-set date's close and scheduled to exit five trading days later.

4.1 Daily capital allocation

On any trading day d with one or more fresh signals, a fixed fraction of the day's starting equity is put at risk:

$$C_d = \phi \cdot E_d, \qquad \phi = 0.55,$$

where E_d is equity (cash plus mark-to-market of open positions) at the start of day d. At most $M = 3$ signals are taken per day, chosen by descending κ, and each is allocated a share of C_d proportional to its own κ relative to the sum of κ across the day's taken signals.

5 Risk Management

5.1 Intraday stop losses

Every open position (other than on its entry day) is checked daily against an intraday stop, referenced to the position's *entry-day opening price* O_0 (not the entry fill price):

$$\text{Long stop: } L_d \le 0.97 \cdot O_0, \qquad \text{Short stop: } H_d \ge 1.03 \cdot O_0,$$

where L_d, H_d are the day's low and high. If triggered, the position is closed at a fixed booked loss of STOP_LOSS_PCT $= -3\%$ of its notional size (rather than the exact intraday execution price), net of a half round-trip transaction cost.

5.2 Transaction costs

A flat round-trip cost of 20 basis points (ROUND_TRIP_COST $= 0.002$) is charged on both entry and exit, applied as half the round-trip cost against notional size at each leg.

5.3 Quiet-day index rotation fallback

On days with no qualifying Markov signal, the strategy does not sit entirely idle. Instead, it ranks a fixed basket of eleven broad index ETFs (SPY, QQQ, IWM, EFA, EEM, EWJ, FXI, EWU, EWG, EWQ, EWZ) by a volatility-adjusted trend score:

$$\text{score}(i) = \frac{\hat{\beta}_i \cdot 252}{\hat{\sigma}_i},$$

where $\hat{\beta}_i$ is the slope of a linear fit to log-price over the trailing 60 trading days, and $\hat{\sigma}_i$ is the trailing 20-day realized volatility, annualized. The strategy goes long the highest-scoring ETF and short the lowest-scoring ETF, sized inversely to their respective volatilities and capped at 25% of equity in aggregate (FALLBACK_BUDGET_FRAC), held for 5 trading days.

6 Backtest Engine

The simulation is a daily event loop over the full business-day range implied by the generated signals. Algorithm 1 summarizes the per-day sequence of operations.

Algorithm 1 Daily backtest loop

```
for each business day d do
    Mark all open positions to market; compute E_d = cash + MTM
    if signals exist for d then
        Open Markov-signal trades, sized per Section 4.1
    else
        Open index-rotation fallback trades (Section 5.3)
    end if
    Apply intraday stop losses to all positions opened on prior days
    Close any position whose scheduled exit day has arrived
    Record end-of-day equity and daily P&L
end for
```

Price data is cached in two forms prior to simulation: an adjusted-close series (for mark-to-market and exit pricing) and a *non-adjusted* OHLC series (needed so that intraday stop checks reference true traded highs/lows rather than dividend/split-adjusted values).

7 Performance Attribution

Let $\{E_d\}$ be the realized daily equity curve and $\{\rho_d\}$ the corresponding simple daily returns. With an assumed annual risk-free rate of 5% ($\rho_f = (1.05)^{1/252} - 1$ daily), the excess return series is $\rho_d^{\text{ex}} = \rho_d - \rho_f$.

Sharpe ratio.

$$\text{Sharpe}_{\text{daily}} = \frac{\overline{\rho^{\text{ex}}}}{\text{sd}(\rho^{\text{ex}})}, \qquad \text{Sharpe}_{\text{annual}} = \sqrt{252} \cdot \text{Sharpe}_{\text{daily}}.$$

Annualized return. Both a compounding-based estimate and a calendar-time CAGR are reported:

$$r_{\text{ann}} = \Big(\prod_d (1 + \rho_d) \Big)^{252/n} - 1, \qquad \text{CAGR} = \Big(\frac{E_{\text{final}}}{E_{\text{initial}}} \Big)^{365.25/\text{days}} - 1.$$

CAPM alpha and beta. The strategy's excess returns are regressed on SPY's excess returns over the same aligned dates:

$$\rho_d^{\text{ex}} = \alpha + \beta \cdot \rho_{d,\text{SPY}}^{\text{ex}} + \epsilon_d,$$

estimated by OLS. Both a linearly annualized alpha (252α) and a compounded annualized alpha ($(1+\alpha)^{252} - 1$) are reported, along with β, R^2, and the t-statistic/p-value on α for a rough read on statistical significance of any outperformance.

8 Discussion and Limitations

- **Survivorship bias.** Index constituents (S&P 500, FTSE 100, Hang Seng, Nikkei 225) are pulled from *current* membership lists and then backtested over 20 years of price history. Firms removed from an index over that period (delisted, acquired, bankrupt) are excluded from the universe, which mechanically biases the backtest's historical performance upward relative to a strategy that could only have known index membership in real time.
- **In-sample edge filter, out-of-sample evaluation.** The edge filter (Section 3) is applied to the training segment; the actual P&L is generated only from test-segment signals. This mitigates but does not eliminate selection effects from testing many tickers simultaneously (multiple-comparisons risk): with a large universe, some tickers will clear a 55% training-set hit-rate threshold by chance alone.
- **Approximated stop-loss execution.** A triggered stop books a fixed -3% loss on notional rather than the exact price at which a gap-through would have executed, which understates losses in the tails (large overnight gaps).
- **No cross-market FX adjustment.** Hang Seng, Shanghai, FTSE, and Nikkei tickers are priced and P&L'd in local currency; no currency-hedging or FX-translation is modeled, so realized USD returns for a US-based investor would differ from what is reported.

- **Data coverage and runtime.** Data availability and quality via `yfinance` varies substantially across exchanges; the volatility-based prefilter (retaining only the top 20% most volatile tickers) is a practical runtime mitigation, not a methodological requirement of the strategy.

This document accompanies the code in the project repository (`markov_strategy/`) and is intended as a methodology reference, not as investment advice. Past backtest performance is not indicative of future results.